

Lab 6: Perceptron Using Scikit-learn

Aim

To implement a **Perceptron classifier using Scikit-learn** for binary classification.

Algorithm

1. Import the required libraries.
2. Create the input dataset and target values.
3. Split the dataset into training and testing sets.
4. Create a Perceptron classifier.
5. Train the model using the training data.
6. Predict the output for the test data.
7. Calculate and display the accuracy.

Program

```
# Lab 6: Perceptron Using Scikit-learn
```

```
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

# Input data
X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])
```

```
# Target output for OR gate
y = np.array([0, 1, 1, 1])

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

# Create and train the model
model = Perceptron(random_state=42)
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Actual Values:", y_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Sample Output

Predictions: [1]

Actual Values: [1]

Accuracy: 1.0

Result

Thus, the **Perceptron classifier** was successfully implemented using **Scikit-learn** and used to perform binary classification.